

BMFLC with Neural Network and DE for better Event Classification

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Abstract—The event-related desynchronization(ERD) is a well known phenomenon that is commonly used for classification in brain-computer interface(BCI) applications. The classification accuracy of ERD based BCI can be improved by selection of subject-specific reactive band rather than complete μ -band. After obtaining time-frequency(TF) mapping of EEG signal with a Fourier based adaptive method, differential evolution(DE) is used for the identification of the reactive band. Compared to classical band-power based method, the proposed method based on subject-specific reactive band yields better accuracy with BCI competition dataset IV.

Keywords—Classification, motor imagery, Optimal Band

I. INTRODUCTION

The brain-computer interface (BCI) is defined as an alternative communication pathway that can directly translate various brain signals into control commands [1]. The sensorimotor ERD is an energy decrease phenomenon that occurs after the subject starts to perform or imagine limbs movement. Due to its robustness, it has been used in various EEG-based BCI applications [2]. For detecting the ERD pattern, a cue-based experiment scheme is generally used. In most subjects the ERD can only be seen in a specific narrow band. It has been shown that the accuracy of BCIs can be improved by employing a narrow subject-specific frequency band [3], [4].

Since the ERD pattern is both time and frequency dependent. Therefore, in order to identify the subject-specific ERD patterns, the time-frequency representation(TFR) methods are generally used [5]. The band-limited Fourier linear combiner(BMFLC) is an adaptive filter based time-frequency decomposition methods that can provide real-time time-frequency mapping of the signal [3].

In [3], [4], the reactive band identification is identified based on visual inspection. The selection of reactive band depends on the maximum energy change in a certain band with respect to a experiment cue [2]. Therefore, the subject-reactive band identification can also be seen as an optimization problem. In order to automatically identify the optimal band for a subject, the differential evolution(DE) algorithm is employed in this paper.

The feature used in the ERD-based BCI systems rely on EEG amplitude in a certain frequency band [6]. The time-frequency mapping that obtained from BMFLC can be directly

used as the feature for the classifier. In this paper, the applicability of the NN with BMFLC is studied for classification of limbs movement.

II. METHOD

To model the unknown μ -rhythm, we consider the signal to be distributed in the band of $[\omega_1 - \omega_n]$ and then employing BMFLC to obtain the time-frequency mapping [3]. The obtained time-frequency decomposition contains the absolute values of all the frequency components with frequency spacing/resolution $\Delta\omega$ and time resolution k . The power for all the frequency components is denoted as $P(k, f)$, where k is the time instant and f is the frequency index. For more details, please see [3], [4].

The DE algorithm [7] finds the optimal of a function $f : R^D \rightarrow R$ by uniform sampling the solution vectors in a D -dimensional space. The solution vectors then go through mutation and crossover in every generation and the fitness solution vector is selected and survived to next generation. After G generations, if the fitness value remains stable then corresponding solution vector is considered as the optimal value.

The objective is to maximize the energy contained in a certain band of the time-frequency mapping obtained with BMFLC. For the selection, we propose a rank strategy to deal with the objective function evaluation problem. Let's denote the number of EEG trials of a movement imagery as n . Thus for each subject and each movement, we have total n time frequency mappings. To compare the fitness of two candidate solutions, we first calculate the ratio of the energy with respect to the total energy in the current TF mapping as follows

$$P_{total}^n = \sum_{f_{min}}^{f_{max}} \sum_{t_{start}}^{t_{end}} P(f, t)^n \quad (1)$$

$$P_{region}^n = \sum_{f_1}^{f_2} \sum_{t_1}^{t_2} P(f, t)^n \quad (2)$$

$$Ratio_n = \frac{P_{region}^n}{P_{total}^n} \quad (3)$$

where $f_{min} = 7$ and $f_{max} = 14$ is selected to cover the whole μ -band. t_{start} and t_{end} is the total length of the trial. f_1 and t_1 are the components of the candidate solution. $f_2 = f_1 + 2$ and $t_2 = t_1 + 2 * F_s$ where F_s is the sampling frequency of the EEG recording. n is the index of the experiment trial. After applying above calculation, our objective is to get higher ratio. For each candidate solution, we have n ratio values. In order to compare two candidate solutions, we employ the following ranking procedure:

$$rank_1 = \begin{cases} 1 & \text{if}(Ratio_n^1 \geq Ratio_n^2) \\ 0 & \text{if}(Ratio_n^1 < Ratio_n^2) \end{cases} \quad (4)$$

$$rank_2 = 1 - rank_1 \quad (5)$$

Thus the $rank$ is a vector of dimension n . For each element, it can only take either 1 or 0. In order to compare two candidate solutions, we take the summation of the $rank$. The candidate solution which results in a high rank will survive to the next generation.

III. RESULTS

The data set used in this report is from BCI Competition IV [8] and contains EEG data of 9 subjects. For hand movement imagery, EEG data from the electrodes C3 and C4, placed over the sensorimotor cortex according to 10/20 international system is selected for analysis. For more details, see [8]. The following parameters are used for modeling EEG with BMFLC [3]:

- **BMFLC:** $f_{min} = 6\text{Hz}$, $f_{max} = 14\text{Hz}$, $\Delta f = 0.2\text{Hz}$
- **DE:** $F = 0.8$, $C_r = 0.7$, $N_p = 20$, $G_{max} = 200$

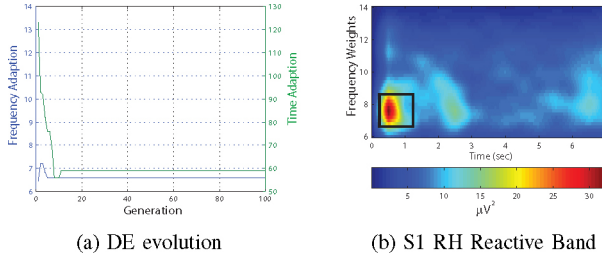


Fig. 1. Reactive band identification with DE

The result obtained with DE based optimal band identification procedure for subject 1 with right hand imagery data is shown in Fig. 1. The DE evolution is shown in Fig. 1(a). It is shown that after less than 20 generations, the fitness value of DE algorithm starts to settle. The obtained reactive region is shown in Fig. 1(b). It is clear the identified reactive band is matching with the TF mapping.

By using the obtained reactive band, the corresponding frequency components are selected from BMFLC TF mapping are fed to a NN. For comparison, the classical band-power (BP) method is employed [4]. The training and testing accuracies are shown in Fig. 2(a). The NN is trained by using Levenberg-Marquardt(LM) algorithm with different number of nodes. It is shown that by employing an optimal band,

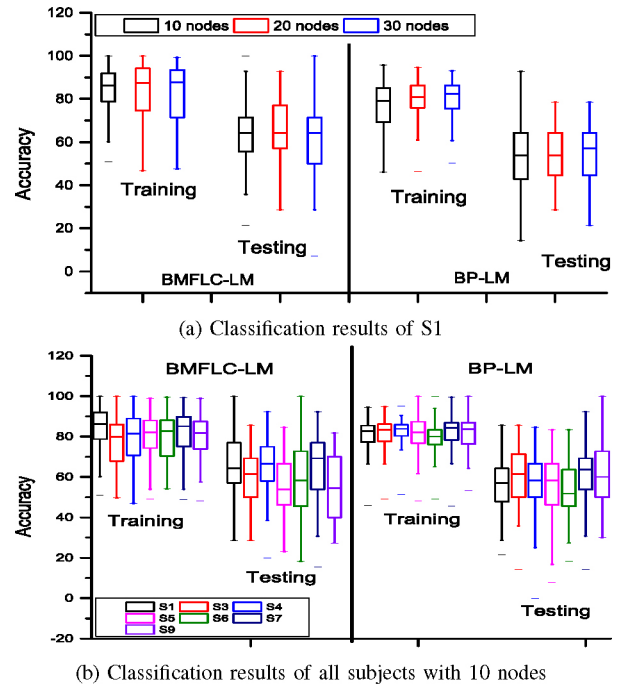


Fig. 2. Classification Results of NN

the training and testing accuracies of BMFLC based classifier are higher than the classical BP based methods. As the 10 nodes NN has lower computational need and provides similar accuracy, we applied this NN to all subjects and the results are shown in Fig. 2(b). It shows that 5 out of 7 subjects shows improved classification accuracy with reactive band and BMFLC for feature extraction.

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